

Risk and Performance Measurement- project

Mateusz Gappa

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1 Introduction

The following report is a work for the course the Risk and performance measurement, Which i have taken at Jagiellonian University The project is divided into three main parts:

1. Data preparation and presentation
2. Basic risk analysis
3. Backtesting analysis

Each section includes executable code that allows the reader to fully reproduce the results if they wish to follow the workflow step by step. add link to the repo

2 Data Preparation and Presentation

Before diving deeper into work, it is essential to establish a clear and transparent foundation for the data we are working with. This chapter focuses on gathering, cleaning, and structuring the data.

We selected a group of 10 companies from the S&P 500 index and examined their performance over the period from 2022-01-01 to 2025-01-01. We used data from the Yfinance python library, (latest version available) This time-frame was proposed by Professor nevertheless it is interesting bi, it captures a dynamic market environment, including post Covid recovery, War at Ukraine and inflationary pressures, and shifting sector trends, all of which make it an interesting window for risk and performance analysis.

2.1 Selected Companies

The process of selecting the companies was non-quantitative in nature. The selection was based on a review of the S&P 500 constituents, ensuring that each chosen company was included in the index throughout the period from 2022-01-01 to 2025-01-01. In addition, care was taken to construct a diversified sample by selecting firms from different industry sectors, thereby reducing concentration risk at the portfolio level. Below table shows selected companies.

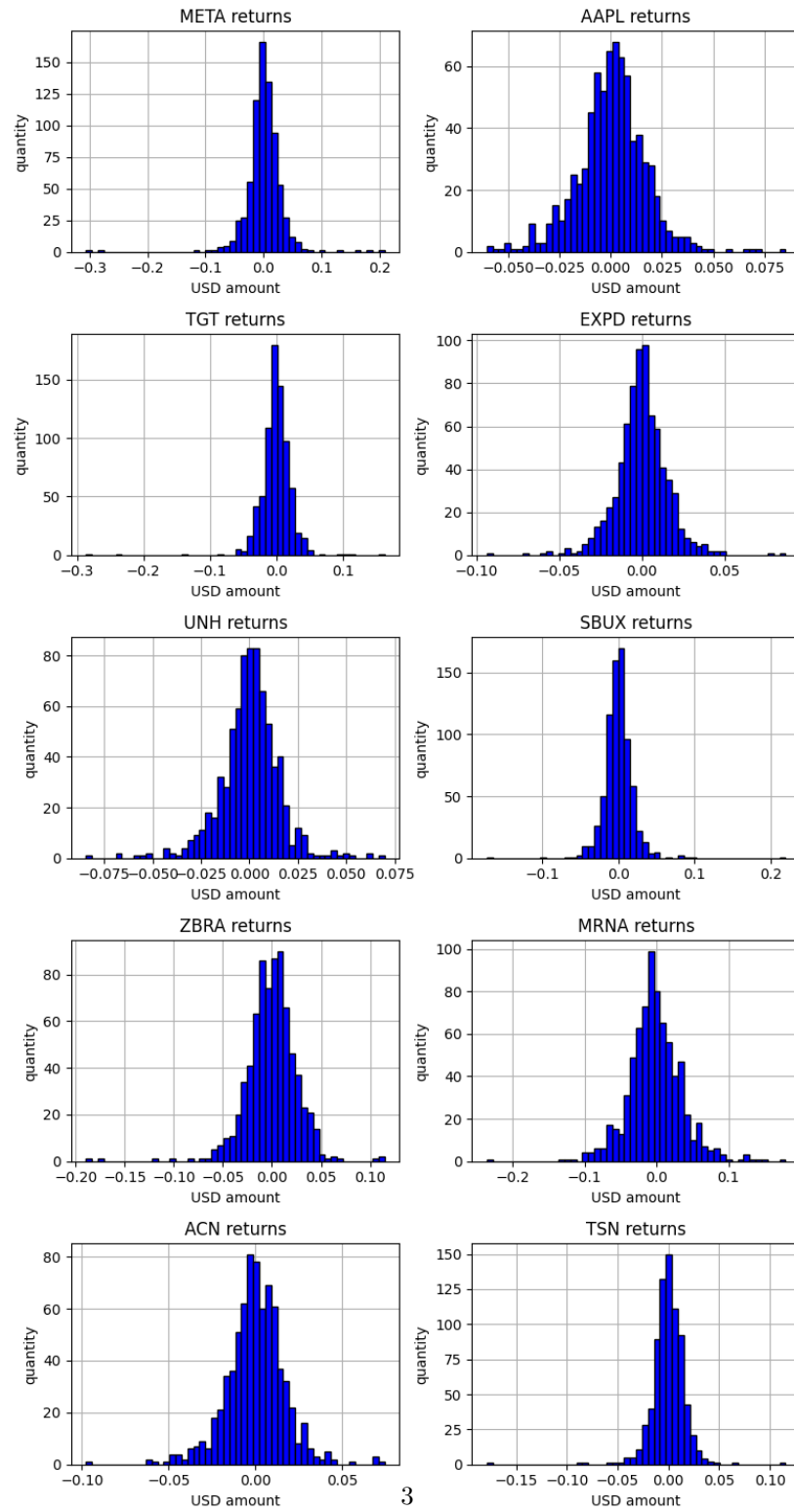
Ticker	Company	Sector
META	Meta Platforms, Inc.	IT
AAPL	Apple Inc.	IT
TGT	Target Corporation	Consumer Discretionary
EXPD	Expeditors International of Washington, Inc.	Industrials (Air Freight & Logistics)
UNH	UnitedHealth Group Incorporated	Healthcare
SBUX	Starbucks Corporation	Consumer Discretionary
ZBRA	Zebra Technologies Corporation	Technology
MRNA	Moderna, Inc.	Healthcare (Biotechnology)
ACN	Accenture plc	IT Services
TSN	Tyson Foods, Inc.	Consumer Staples

Table 1: Selected S&P 500 companies

2.2 Distribution of Daily Returns

To better understand the behavior of individual assets, we visualized the distribution of daily returns for each company using histograms. In addition, time-series plots of returns were briefly analyzed; however, as no notable patterns or anomalies were observed, the analysis was limited to the distributional perspective provided by the histograms.

Most of the distributions on the histograms appear approximately normal and are centered around zero, which is typical for daily returns in liquid markets. However, some assets exhibit heavier tails, such as UNH, or slight skewness, as observed for AAPL. This indicates the presence of occasional large price movements or asymmetric return behavior. But once again doing visual inspection we didn't noticed anything suspicious thus there is no need to perform further formal analysis, such as volatility clustering tests.



3

Figure 1: Histograms of the returns

To extend the analysis, we wanted to look at the statistical components such as Mean and variance.

Ticker	Mean	Variance
Zebra Technologies	-0.00023	0.00064
Moderna, Inc.	-0.00158	0.00145
Meta Platforms, Inc.	0.00121	0.00094
Starbucks Corporation	-0.00003	0.00042
Accenture plc	0.00001	0.00029
Expeditors International	-0.00005	0.00025
UnitedHealth Group	0.00019	0.00024
Apple Inc.	0.00059	0.00029
Tyson Foods	-0.00031	0.00026
Target Corporation	-0.00030	0.00059

Table 2: Empirical mean and variance of daily returns

The empirical mean and variance estimates of daily returns indicate a generally unfavorable market environment, with only Apple Inc. and Meta Platforms, Inc. exhibiting slightly positive average returns, while the remaining assets show negative means. It can be explained macroeconomically. Variance which can be seen as "Risk" varies across companies, with Moderna, Inc. and Meta Platforms, Inc. displaying the highest variances, and UnitedHealth Group, Expeditors International, and Accenture plc characterized by comparatively lower volatility.

The empirical mean and variance estimates of daily simple returns presented in Table ?? indicate a challenging market environment over the analyzed period. Only Apple Inc., Meta Platforms, Inc., UnitedHealth Group, and Accenture plc exhibit marginally positive average returns, while the remaining companies display negative mean returns, suggesting limited profitability at the individual asset level. Variance which can be seen as "Risk" varies across companies, with Moderna, Inc. and Meta Platforms, Inc. displaying the highest variances, and UnitedHealth Group, Expeditors International, and Accenture plc characterized by comparatively lower volatility.

2.3 Correlation Analysis

The correlation heatmap presents pairwise correlations between daily closing prices of the selected assets. Most assets exhibit moderate levels of correlation, particularly within the technology and consumer-related sectors. Apple and Accenture show the strongest relationship, with a correlation coefficient of 0.56, while Moderna and UnitedHealth display relatively low correlations, indicating potential diversification benefits.

This correlation structure will be useful in subsequent sections when analyzing portfolio risk and asset allocation.

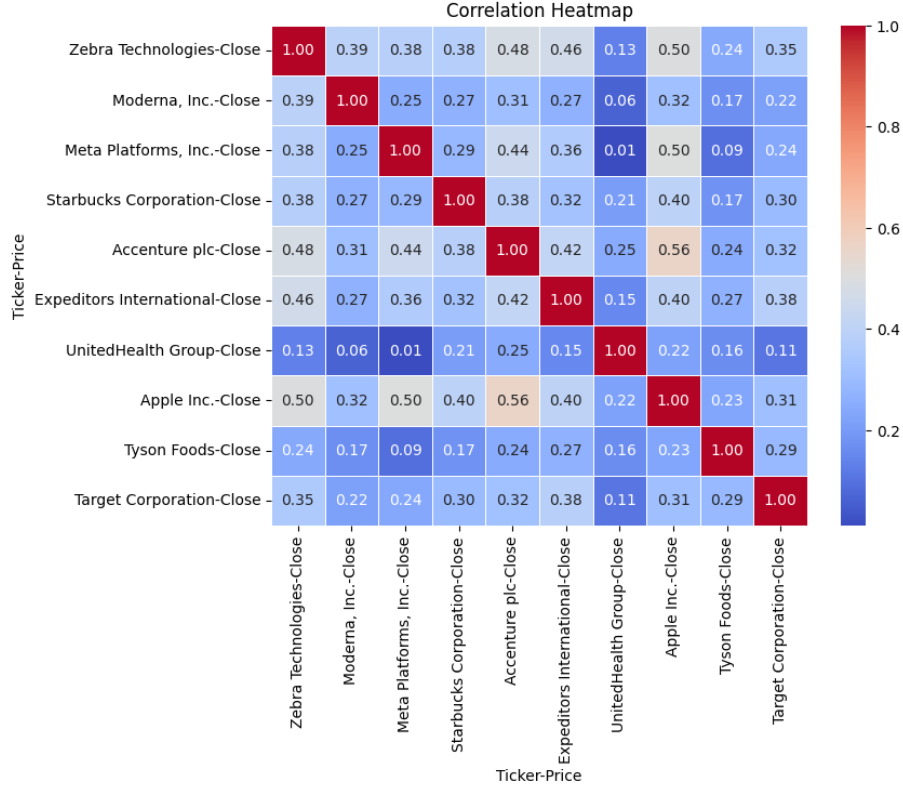


Figure 2: Correlation Plot

3 Basic Risk Analysis

The objective of this section is to provide an initial quantitative assessment of risk, allowing for a comparative evaluation of assets before proceeding to portfolio-level analysis and backtesting.

3.1 Risk-Adjusted Performance and Drawdown Measures

In this section we have to show two additional metric of risk which were used in analysis.

First one is a **Sharpe ratio** defined as:

$$SharpeRatio = \frac{E[R_p] - R_f}{\sigma(R_p)} \quad (1)$$

where:

- R_p denotes the expected return of the portfolio,

- R_f denotes the risk-free rate,
- σ_p denotes the standard deviation of the portfolio returns.

Additional metric that i chosed to use is **Maximum Drawdown**

$$MaximumDrawdown = \min_{t \in [0, T]} \left(\frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s} \right) \quad (2)$$

where:

- V_t denotes the value of the portfolio (or asset) at time t ,

Going foward i will denote Sharpe Ratio as SR dn Maximum Drawdown as MDD

The reason for choosing the Sharpe ratio as the first additional metric is its simplicity and the overview which it gives. I am interpreting this metric as indicating how much is earned above the risk-free rate and how volatile the strategy is. This simplicity and the overview it provides were the reasons why I chose this measure.

The reason why I chose MDD is completely psychological. This metric basically gives the percentage loss experienced at the worst moment. Some might say that this measure is too conservative, but I believe that in the real world of financial markets, controlling one's psyche is crucial.

3.2 Results

Company	Mean (daily)	Std (USD)	VaR 1% (USD)	ES 2.5% (USD)	Sharpe ratio	Max Drawdown (USD)
Zebra Technologies	-0.0029	58.6	144.6	162.2	-0.0975	-1227.8
Moderna, Inc.	-0.0001	91.2	183.6	182.6	-0.0012	-997.6
Meta Platforms, Inc.	-0.0033	81.2	171.8	257.4	-0.0804	-1471.6
Starbucks Corporation	-0.0003	45.2	101.8	100.4	-0.0133	-767.2
Accenture plc	-0.0014	41.8	108.6	104.0	-0.0679	-746.0
Expeditors International	-0.0007	39.8	102.0	105.6	-0.0340	-657.4
UnitedHealth Group	0.0004	30.8	73.4	71.0	0.0250	-338.2
Apple Inc.	-0.0011	45.0	107.6	103.2	-0.0478	-589.0
Tyson Foods	-0.0011	34.4	92.0	98.6	-0.0662	-744.8
Target Corporation	-0.0012	58.2	114.6	186.8	-0.0424	-878.0

Table 3: Empirical risk and performance measures based on daily returns in 2022 for an investment of 2000 USD

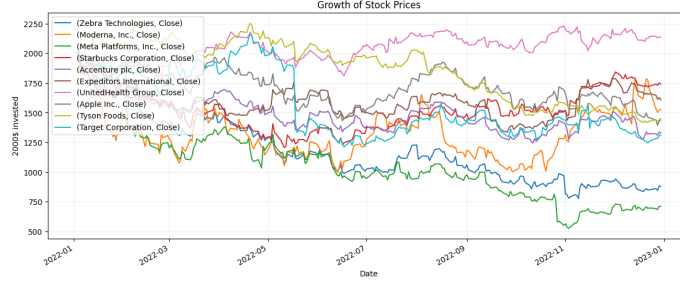


Figure 3: 2000 USD invested into each company at 2022

3.3 Portfolio Definitions

Three investment scenarios are considered:

1. **Top-3 stock portfolio (2000 USD).** The full budget of 2000 USD is invested in three stocks selected based on the empirical results obtained in the previous section. The selected stocks are UnitedHealth Group, Starbucks Corporation, and Expeditors International.
UnitedHealth Group was chosen as it was the only stock with a positive Sharpe ratio over the analyzed period, indicating the most favorable risk-adjusted performance in 2022. Starbucks Corporation and Expeditors International were included due to their relatively low volatility and Sharpe ratios close to zero, and their MDD is below 1000usd.
2. **Excluding-top-3 portfolio (2000 USD).** The same 2000 USD is invested across seven stocks, explicitly excluding the top-3 selections from the previous case.
3. **Optimized 5-stock portfolio (inverse Expected Shortfall).** The third investment strategy is based on a risk-oriented portfolio construction using five selected stocks. Unlike the previous portfolios with equal-weight allocations, this strategy applies a simple heuristic optimization rule, where portfolio weights are set inversely proportional to the empirical Expected Shortfall at the 2.5% level.

3.4 Portfolio Results

In this section, we present the results of the 3 portfolios simulations conducted for the year 2022.

Table 4: Empirical risk and performance measures for three portfolio strategies in 2022 (investment of 2000 USD).

Strategy	Mean (daily)	Std (USD)	VaR 1% (USD)	ES 2.5% (USD)	Sharpe (ann.)	Max Drawdown (USD)
Top-3 (equal)	-0.000197	30.256542	61.439454	67.836096	-0.331978	-475.953182
Exclude-3 (equal)	-0.001576	43.151257	107.636932	106.960509	-1.245830	-822.107846
Optimized-5 (inv ES)	-0.000648	32.334974	73.491891	75.088784	-0.751052	-536.847443

Table above compares the empirical risk and performance characteristics of three portfolio strategies assuming an initial investment of 2000 USD. The Top-3 equally weighted portfolio exhibits the smallest downside risk, as reflected by the lowest Value at Risk and Expected Shortfall, as well as the shallowest maximum drawdown. In contrast, the Exclude-3 portfolio shows the weakest performance, characterized by the most negative mean return, the highest volatility, and the largest MDD, resulting in a strongly negative Sharpe ratio. The Optimized-5 portfolio, constructed using inverse Expected Shortfall weights, achieves an intermediate outcome. As we noticed before year 2022 was noticable bad for financial markets thus we are not amazed by poor results of our strategies

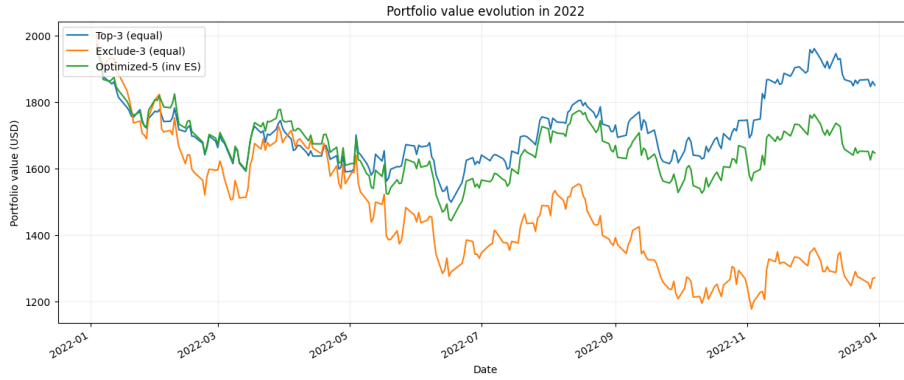


Figure 4: Portfolio value evolution in 2022

As indicated by the tabular analysis, all considered portfolio strategies resulted in losses over the analyzed period, in line with the performance ranking discussed earlier.

4 backtesting

In this section, we perform a backtesting analysis of the third portfolio over the years 2023-2024. We begin by evaluating the historical Value at Risk at the 1% confidence level, computed using daily simple returns and a rolling one-year lookback window. The results are then directly compared with a benchmark portfolio composed of the same five stocks, but constructed under an equal-weight allocation scheme.

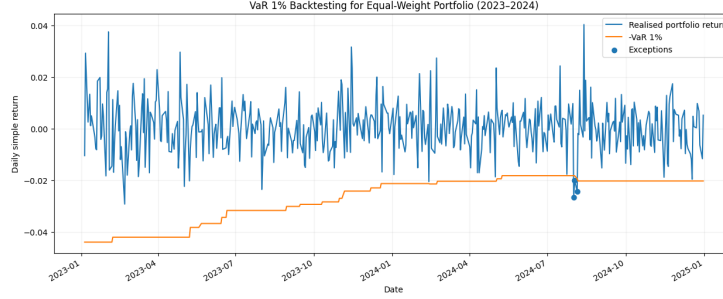


Figure 5: VaR 1% backtesting for equal-weight portfolio

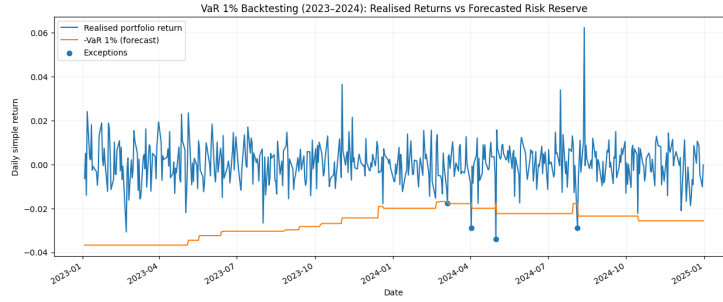


Figure 6: VaR 1% backtesting for Optimized-5 portfolio

We begin by analyzing the Optimized-5 portfolio. In the backtesting period, a total of four VaR exceptions (overshoots) are observed, indicating that the estimated 1% Value at Risk provides an adequate level of risk protection for the portfolio. Under the assumption that daily returns follow the true underlying distribution, the number of VaR exceptions should follow a binomial distribution $\mathcal{B}(n, 0.01)$, where n denotes the number of trading days. In our case, $n = 500$.

According to the supervisory traffic light framework introduced by the Basel Committee on Banking Supervision Basel 1996, the 95% quantile of the binomial distribution $\mathcal{B}(501, 0.01)$ equals 9, this marks the threshold between the green and yellow zones. Since the observed number of exceptions remains well below this limit, the backtesting results fall within the green zone.

4.1 Final comparison

To compare the performance of the optimized portfolio with a simple benchmark, we evaluate both strategies using realized returns over the out-of-sample period from January 2024 to the end of November 2025. The analysis focuses on standard performance and risk measures, including the mean and standard deviation of daily returns, Sharpe ratio, and the final portfolio value assuming an initial investment of 2000 USD.

Strategy	Mean	Std	Sharpe	Final value (USD)
Equal-weight (5)	0.0005	0.0096	0.6321	2243.92
Optimized-5 (inv ES)	0.0004	0.0095	0.4511	2181.45

Table 5: Realized performance measures for the equal-weight and Optimized-5 portfolios over the period 2024–2025 (up to November 2025), assuming an initial investment of 2000 USD.

It might be surprising but the equal-weight portfolio outperforms the optimized allocation in terms of both realized mean return and risk-adjusted performance, as reflected by a higher Sharpe ratio and a larger final portfolio value. This result suggests that the tail-risk-based optimization calibrated on 2022 data does not generalize well to the subsequent period. In particular, changes in market conditions and sectoral return dynamics in 2024-2025. The reason for that might be, as mentioned above, changes in the global political situation, and also unpredictable AI boom in 2024, which is in favour of companies in the IT sector.